



EE313 - ELECTRONICS II
HOMEWORK 2 REPORT

Student: Elif Sude ÖZTÜRK

Number: 310206045

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1) Introduction:

The purpose of this assignment is to design and analyze three separate continuous-time LTI systems that can individually extract specific frequency components from a composite input signal. The given input signal consists of three sinusoidal components at 500 Hz, 2 kHz, and 10 kHz as below.

$$y_1(t) \approx \sin(2\pi f_1 t),$$

$$y_2(t) \approx \sin(2\pi f_2 t),$$

$$y_3(t) \approx \sin(2\pi f_3 t).$$

For achieving to this goal we developed a system that shown in **Figure 1**. Each system is designed to pass one desired frequency component while sufficiently attenuating the remaining two components. To achieve this objective, appropriate active filter structures are selected based on frequency-domain characteristics.

In this study, low-pass, band-pass, and high-pass filter topologies are employed to isolate the low-, mid-, and high-frequency components of the signal, respectively. Analytical calculations are performed to determine suitable cutoff frequencies and filter orders, and the designs are verified using LTspice.

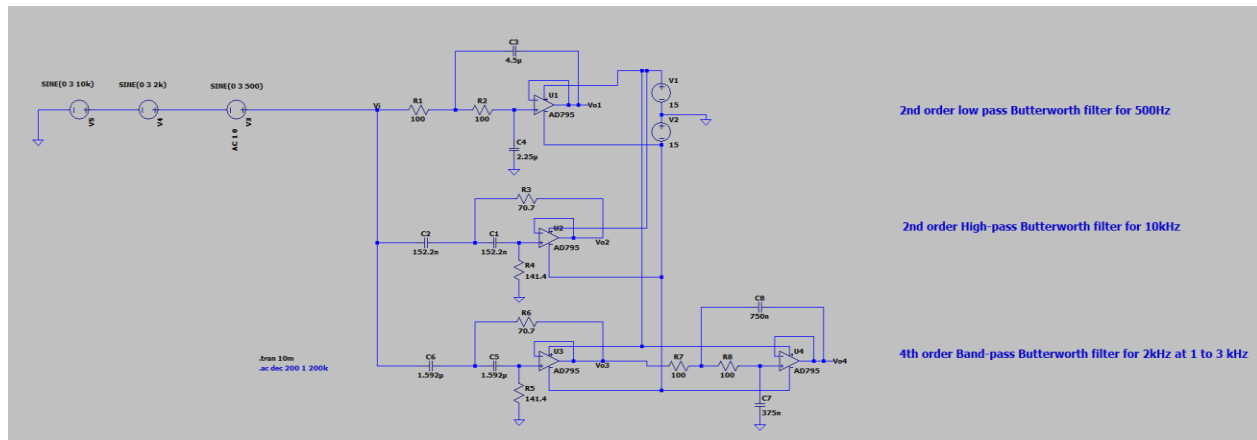


Figure 1: The Filter System

2)Theoretical Background and Design

We used Butterworth filters for the filtering process. The reason we chose this filter type is that Butterworth filters are considered the most flat filters. For this purpose, we used the general two-pole active filter topology as shown in **Figure 2**. The parts indicated with Y are admittance. We are concerned with the transfer functions of the circuits for the filtering process. The transfer function of this topology is as shown in **Figure 3**. We are going to use this topology to develop low-pass, high-pass, and band-pass filters.

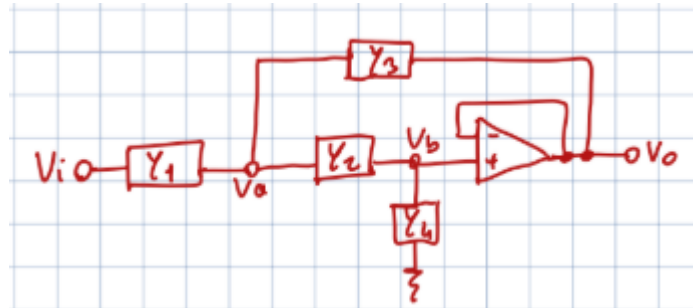


Figure 2: 2 Pole Active Filter Topology

$$T(s) = \frac{V_o(s)}{V_i(s)} = \frac{Y_1 Y_2}{Y_1 Y_2 + Y_4 (Y_1 + Y_2 + Y_3)}$$

Figure 3: Transfer Function of General 2 Pole Active Filter

2.1) Low-Pass Butterworth Filter:

To obtain a low-pass Butterworth filter, the admittances Y3 and Y4 are replaced with capacitive elements, while equal resistances are placed at the Y1 and Y2 locations. At high frequencies ($\omega \rightarrow \infty$) the capacitors behave as short circuits, resulting in attenuation at the output. Conversely, at low frequencies ($\omega \rightarrow 0$) the capacitors act as open circuits, allowing the input signal to pass to the output with approximately unity gain. Following the general two-pole active filter formulation, a second-order low-pass Butterworth filter is designed to extract the 500 Hz component of the input signal. Their calculations and simulation outputs are below.

Assumption $\rightarrow G_1 = G_2 = G = \frac{1}{R}$ (admittances)

$$T(s) = \frac{\frac{1}{R^2}}{\frac{1}{R^2} + sC_4\left(\frac{2}{R} + sC_3\right)} = \frac{1}{1 + sRC_4(2 + sRC_3)}$$

$s = j\omega$

$$|T(j\omega)| = \frac{1}{\sqrt{1 + 4(\omega RC_4)^4}}$$

$$\left. \frac{d|T|}{d\omega} \right|_{\omega=0} = 0 \rightarrow C_3 = 2C_4$$

$$\omega_{\text{dB}} = 2\pi f_{\text{dB}} = \frac{1}{\sqrt{2}RC_4}$$

For $f_1 = 500 \text{ Hz}$, we selected the cut-off frequency as $f_c = 500 \text{ Hz}$ for attenuate the $f_2 = 2 \text{ kHz}$ and $f_3 = 10 \text{ kHz}$.

$$f_c = \frac{1}{2\sqrt{2}\pi RC_4} = 500 \text{ Hz.} \quad RC_4 = 2.251 \times 10^{-4} \quad \text{Let, } R = 100 \Omega \rightarrow C_4 = 2.251 \mu\text{F.}$$

$$C_3 = 4.5 \mu\text{F.}$$

Figure 4: Calculations

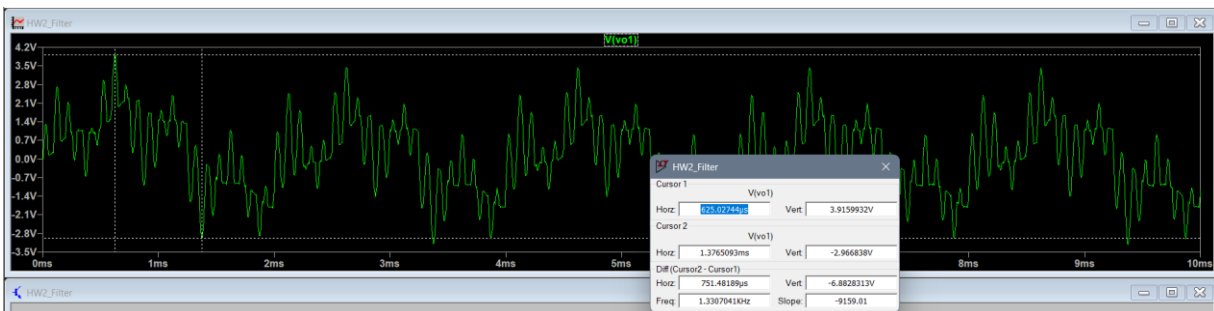


Figure 5: Voltage-Time graph of Low-Pass Butterworth Filter

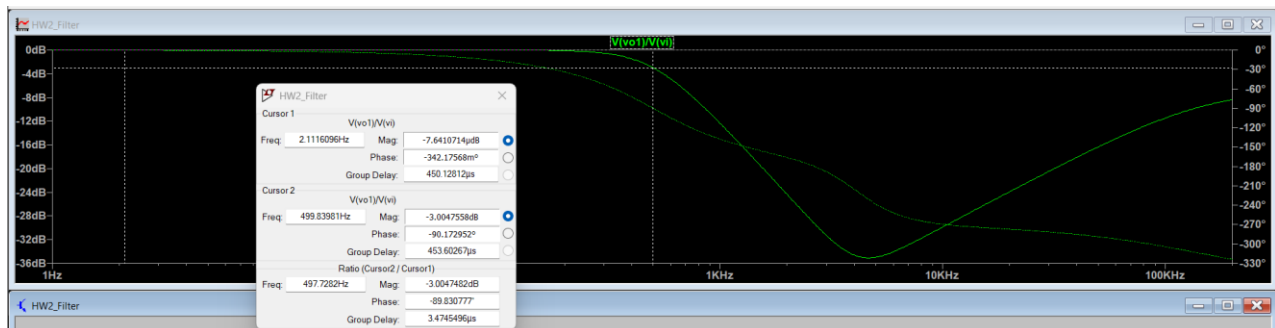


Figure 6: Bode Plot of Low-Pass Butterworth Filter Transfer Function

As shown in **Figure 6**, our filter successfully suppresses frequencies above 500 Hz. This also confirms that our calculations in **Figure 4** are correct. In **Figure 5**, we see a sinusoidal shape but with regular distortion. This is caused by our other 2KHz and 10KHz signals. Our filter does not completely eliminate other signals; it suppresses them. Therefore, we do not encounter a clean sine wave at the filter's output. However, if we want a smoother output, increasing the filter's order will suffice.

2.2) High-Pass Butterworth Filter:

To obtain a high-pass Butterworth filter, the admittances Y_1 and Y_2 are replaced with resistance elements, while equal capacitive elements are placed at the Y_3 and Y_4 locations. At high frequencies ($\omega \rightarrow \infty$) the capacitors behave as short circuits, allowing the input signal to pass to the output with approximately unity gain. Conversely, at low frequencies ($\omega \rightarrow 0$) the capacitors act as open circuits, resulting in attenuation at the output. Following the general two-pole active filter formulation, a second-order high-pass Butterworth filter is designed to extract the 10KHz component of the input signal. Their calculations and simulation outputs are below.

Y_1 and Y_2 replaced with same C capacitor. Y_3 and Y_4 replaced with R_3 and R_4 .

$$|T| = \frac{1}{\sqrt{1 + \left(\frac{f_{3dB}}{f}\right)^4}} \quad \omega_{3dB} = 2\pi f_{3dB} = \frac{1}{RC} \quad R_3 = 0.707R \quad R_4 = 1.414R$$

for $f_3 = 10\text{kHz}$, we decide our cut-off frequency is 10kHz .
Because of the filter is high-pass f_1 and f_2 frequencies are attenuated.

$$f_c = \frac{1}{2\pi RC} = 10\text{kHz} \quad RC = 1.592 \times 10^{-5} \quad \text{if } R = 100\Omega, C = 159.2\text{ nF.}$$

$$R_3 = 70.7\Omega \text{ and } R_4 = 141.4\Omega$$

Figure 7: Calculations for High-Pass Butterworth Filter

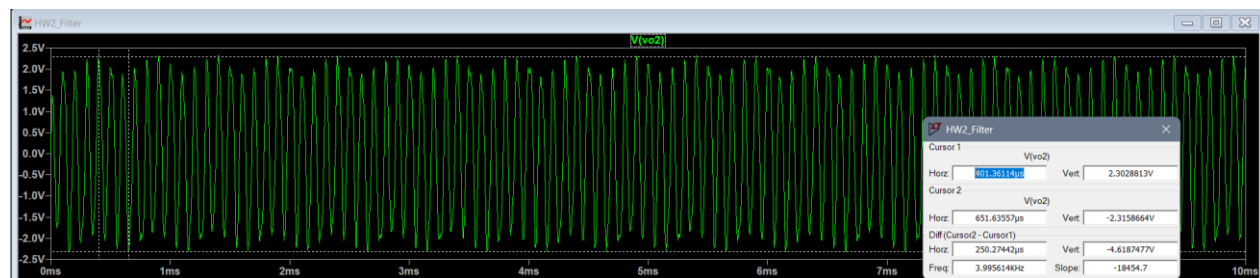


Figure 8: Voltage-Time graph of High-Pass Butterworth Filter

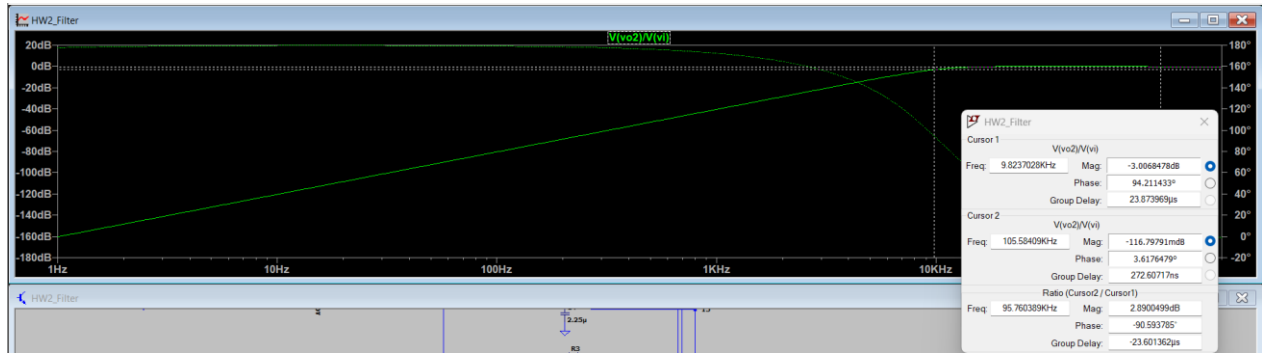


Figure 9: Bode Plot of High-Pass Butterworth Filter Transfer Function

As shown in **Figure 9**, all frequencies below 10 kHz have been successfully suppressed. This also confirms that our calculations in **Figure 7** are correct. **Figure 8** shows the voltage-time graph of the filtered signal. Here we can see that the 10 kHz signal has been successfully passed. The shape we see is almost a pure sinusoidal shape, but unfortunately, as we mentioned in low-pass Butterworth filters, there are also parts from the other two signals. This causes distortion here. For a cleaner output, we need to design a higher-order filter.

2.3) Band-Pass Butterworth Filter:

To obtain band-pass butterworth filter, we connected one high-pass butterworth filter and one low-pass butterworth filter in cascaded format. This makes our band-pass filter 4th order band-pass. Our aim in this design is to pass 2KHz signal and attenuate the other two signal. We decide our band is 1KHz to 3KHz. High-pass filter is going to pass the signals greater than 1KHz, and low-pass filter is going to pass the signals less than 3KHz. The calculations and simulation outputs are below.

for cut-off frequency 1kHz high-pass, 3kHz lowpass.

$$f_{c,L} = \frac{1}{252 \pi R C_4} \approx 3 \text{ kHz}$$

$$R C_4 = 3.75 \times 10^{-5}$$

$$\downarrow \begin{matrix} 100 \Omega \\ \rightarrow 0.375 \mu\text{F} \end{matrix} \quad C_3 = 0.75 \mu\text{F}$$

$$f_{c,H} = \frac{1}{2 R C \pi} = 1 \text{ kHz}$$

$$R C = 1.592 \times 10^{-4}$$

$$\downarrow \begin{matrix} 100 \Omega \\ \rightarrow 1.592 \mu\text{F} \end{matrix} \quad R_3 = 70.7 \Omega$$

$$R_4 = 141.4 \Omega$$

Figure 10: Calculations

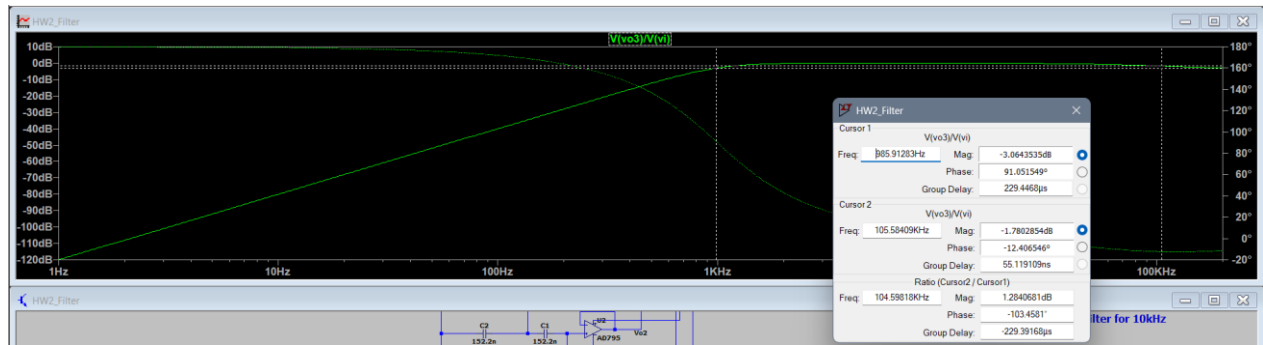


Figure 11: Bode plot of High-Pass Filter Transfer Function for 1KHz

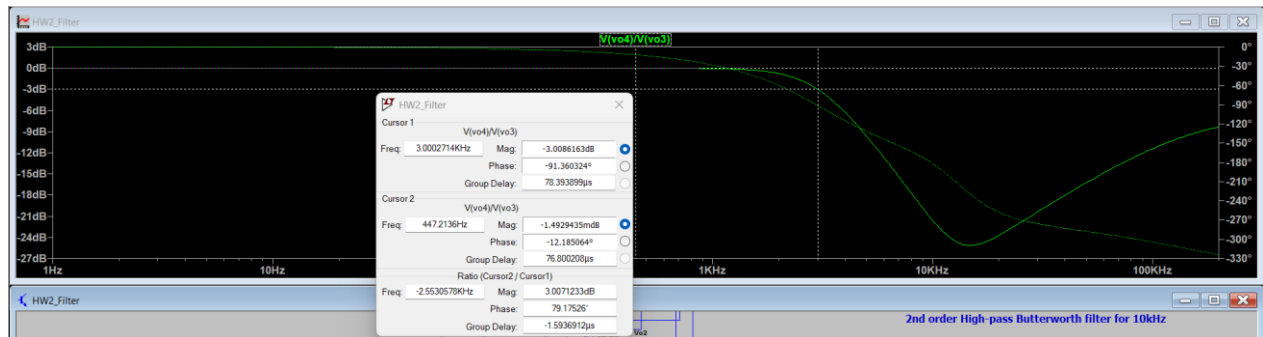


Figure 12: Bode plot of Low-Pass Filter Transfer Function for 3KHz

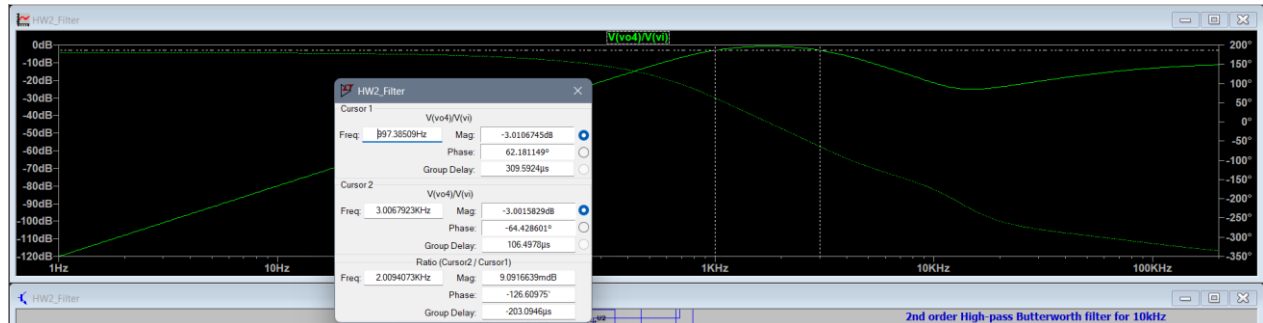


Figure 13: Bode plot of Band-Pass Filter Transfer Function

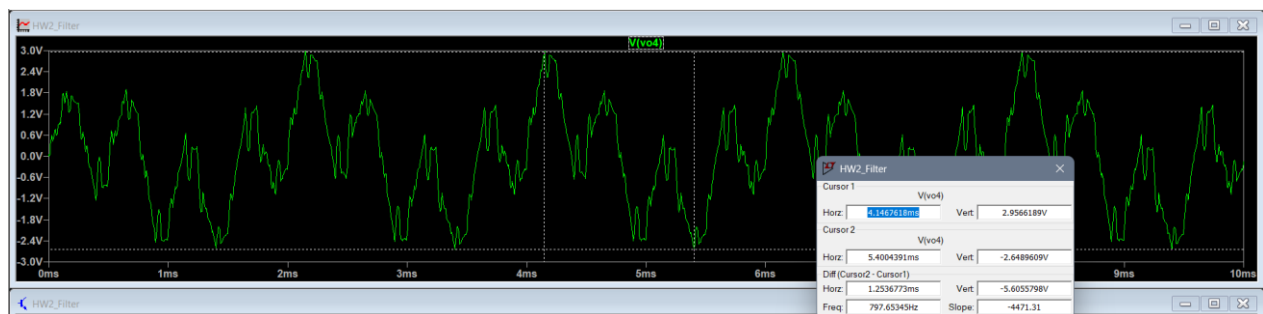


Figure 14: Voltage Time graph of Band-Pass Butterworth Filter

As shown in **Figure 11**, the first cascaded filter passes signal values at 1 KHz and above. **Figure 12** shows that the signal coming from the first cascaded filter passes frequencies of 3KHz and below. Thanks to these two, we have obtained a band pass filter that passes signals with frequencies between 1 and 3KHz. Its transfer function is shown in **Figure 13**, and as we said, it successfully passes frequencies between 1 and 3KHz. In **Figure 14**, we see the voltage-time response of the band-pass filter system. Distortions have occurred in this graph, originating from signals between 500Hz and 10KHz. To achieve smoother and sharper filtering, we need to increase the order of the Butterworth filter we have set up.

3) Conclusion:

In this homework, three separate continuous-time LTI systems were designed to extract individual frequency components from a composite signal consisting of 500 Hz, 2 kHz, and 10 kHz sinusoidal signals. To achieve this objective, low-pass, high-pass, and band-pass Butterworth filters were implemented using active filter topologies.

The filter designs were first developed analytically based on the general two-pole active filter formulation and Butterworth filter conditions. Appropriate cutoff frequencies and filter orders were selected to ensure sufficient attenuation of undesired frequency components while preserving the desired signal. The theoretical designs were then verified through LTspice simulations using both frequency-domain and time-domain analyses.

Simulation results demonstrate that each designed filter successfully passes its intended frequency component while suppressing the remaining components. It was also observed that due to the finite filter order, undesired signals were attenuated but not completely eliminated, resulting in slight distortions in the output waveforms. Increasing the filter order would improve frequency selectivity and produce smoother output signals.

Overall, this study confirms that Butterworth active filters provide an effective and practical solution for frequency separation problems, and that analytical design methods are in strong agreement with simulation results under ideal assumptions.